Calibrated Hallucination Risk for Large Language Models Under Domain and Task Shift with Evidence-Consistent XGBoost

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Abstract

Large language models answer fluently, but some answers are not supported by the evidence. Detectors that rely on a large judge model, an encoder, or model internals are costly and rarely report calibration under domain or task shift. Feature-based detectors are cheaper, but their calibration and sensitivity to answer length are seldom measured together. This study asks whether a simple feature model can estimate hallucination risk and how its ranking and calibration change under shift. We present EC-XGB, an evidence-consistent XGBoost model that combines base features, claim-level natural-language-inference aggregates, and a source indicator. The model is trained on HaluEval QA and RAGTruth non-QA rows with grouped splits and three seeds, and evaluated with fixed-hyperparameter ablations, source and target calibration, conformal prediction, SHAP stability, a length-only control, and judge stress tests. A feature model is cheap and explainable, and its inputs can be inspected directly, which makes it a practical option for routine screening. On HaluEval QA the fixed-hyperparameter model reaches F1 0.985 and area under the receiver operating characteristic curve (AUROC) 0.999, but lexical overlap and length carry that score. On MedHallu it ranks the hallucinated answer above the grounded one in 86.7% of pairs, but it flags every row and is badly calibrated (expected calibration error 0.493). On RAGTruth QA it stays near chance. Target calibration lowers ECE from 0.731 to 0.125 but flags no target rows at the default threshold, and split conformal reaches 0.944 coverage at 90% with only 0.685 positive coverage. Dialogue is flagged almost entirely, summarization ranks below a length-only model, and a GPT judge beats the feature model on natural QA. EC-XGB is therefore a domain-specific risk score, not a general-purpose detector.

1. Introduction

Black-box detectors score a candidate answer from its text and context. SelfCheckGPT samples several answers and checks whether they agree [1]. Luna replaces the large judge with a smaller evaluation model [2]. Semantic entropy clusters sampled answers by meaning [3]. A hybrid framework combines frozen encoder features with lightweight classifiers [4]. Judges carry high computational costs, and encoder models are hard to interpret. Two recent reviews group detectors into feature-based, encoder-based, and hybrid approaches [5, 6]. Both name evaluation and reliability as open problems, and a third review reaches the same conclusion about reliability [7].

HaluAgent showed that small models can select tools, but it needs multiple calls per answer [8]. Valentin et al. proposed a calibrated multi-score detector with a cost analysis, but it relies on LLM scoring [9]. Sundaragiri et al. combined retrieval, NLI, and XGBoost, but report only classification metrics [10]. Haq et al. assigned SHAP-LIME attributions to each word, but did not measure their reliability [11]. Deng et al. found that detectors degrade outside their training domain, but focus on multi-turn dialogue [12]. FaithBench shows that modern LLMs still produce summarization hallucinations, but does not measure detector calibration [13].

The existing literature leaves four issues unaddressed. Calibration is infrequently reported for lightweight detectors, explanation quality is rarely measured, cross-domain behavior is rarely tested, and error patterns are rarely analyzed case by case. The conventional feature-based baseline also has limits. It depends on answer length, it has no claim-level evidence, it trains on HaluEval alone with a source-fitted calibrator, and its probability does not transfer to a new corpus [14]. HaluEval is also saturated. The gap is a risk score that keeps its ranking and probability when the corpus changes. This paper addresses these gaps with a feature-based model that scores the output of a black-box LLM. It takes a query q, a reference context c, and a candidate answer a, and outputs $P(\text{hallucinated} \mid q, c, a)$. EC-XGB combines 26 base features in seven groups, eight claim-level NLI aggregates, and a source indicator in a tuned XGBoost classifier. Platt

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calibration maps the score to a probability, SHAP explains feature contributions, and grouped splits with length and overlap controls reduce shortcut solutions. Source calibration is fitted on validation rows only, and the target-calibration experiments measure the reliability problem. The log transform compresses the length scale but does not remove the cue. The claim-level aggregates support per-claim explanations, and the measured seen-task gain comes from adding multi-source rows and the source indicator, not from the claim aggregates alone. Transfer limits are established on RAGTruth and FaithBench. The evaluation adds MedHallu [15], a HaluBench subset [16] with no training overlap, the HaluEval task shifts, conformal prediction, and judge stress tests. It reports ranking, calibration, and paired errors rather than F1 alone. The added evidence separates ranking from calibration: the model transfers its ranking to medical pairs but not its probability, and it fails under a task shift.

HaluEval QA provides 10,000 question pairs, each with a correct and a hallucinated answer, for training and in-domain testing. RAGTruth provides natural responses for adaptation and recalibration. The target use case is question answering with a reference context, and FaithBench serves only as an out-of-domain summarization stress test. The fixed-hyperparameter EC-XGB stack (m3) reaches F1 0.9854 and AUROC 0.9986 on HaluEval QA. RAGTruth shows the same pattern: the flag rate falls on the tasks seen in training, while held-out QA stays near chance. On MedHallu the same model ranks the paired hallucinated answer above the grounded one in 86.7% of 1,000 pairs but flags every row. On HaluEval dialogue it flags almost every row, and on HaluEval summarization it ranks below a length-only model. Target calibration on 5,034 RAGTruth QA calibration rows lowers ECE from 0.731 to 0.125, but it does not change the ranking. One answer takes 127.5 ms on GPU (2.00 s on CPU) at about $0.001 per 1,000 predictions.

2. Conventional Method

2.1. Preliminaries

This section describes the conventional feature-based detector and the criteria used to evaluate it. A hallucination is an answer that the reference context does not support, or that the context contradicts. Each row is a triple (q_i, c_i, a_i) of question, reference context, and candidate answer, with label $y_i \in \{0, 1\}$. A detector is usually scored with precision, recall, F1, or the area under the receiver operating characteristic curve (AUROC) on labeled rows. A deployed detector faces two more demands. Calibration requires the predicted risk to match the observed frequency, and expected calibration error (ECE) summarizes the gap across confidence bins [17]. Explanation requires the warning to name the features that contributed to it, and Shapley additive explanations (SHAP) provide that for a tabular model [18]. This work measures both properties on the black-box model.

2.2. Baseline Method

The conventional method is a tuned XGBoost classifier [19] over the 26 base features in Table 1, trained on HaluEval QA. XGBoost models nonlinear interactions and gives exact TreeSHAP values, which the explanation analysis needs. A published feature-based detector uses the same learner with retrieval and NLI features [10], so this model is the natural baseline. Trees are added one at a time, and each tree fits the current gradient and curvature of the loss. A row enters at the root of each tree, follows the split conditions to one leaf per tree, and the leaf weights are summed into a raw margin. A logistic link maps the raw margin to a probability.

Let $x_i \in \mathbb{R}^{26}$ be the feature vector of row i, $y_i \in \{0, 1\}$ its label, and n the number of training rows. The model sums K trees into a raw margin, $z_i = \sum_{k=1}^{K} f_k(x_i)$. The training objective combines a differentiable loss with a regularizer over the trees,

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, z_i) + \sum_{k=1}^{K} \left(\gamma T_k + \tfrac{1}{2}\lambda \|w_k\|^2\right), \tag{1}$$

where $\phi = \{f_k\}_{k=1}^{K}$ collects the trees, f_k is the k-th tree with T_k leaves and leaf-weight vector $w_k \in \mathbb{R}^{T_k}$, γ is the penalty per leaf, and λ is an L2 weight penalty. The loss $l(y_i, z_i)$ is the logistic loss on the raw margin. The probability is the sigmoid of the raw margin, $\hat{p}_i = \sigma(z_i) = 1/(1 + e^{-z_i})$.

At boosting step t, a new tree f_t with T leaves and leaf weights $w_1, \ldots, w_T$ is added, and the objective is approximated, up to a constant, by a second-order expansion around the current raw margin,

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \tfrac{1}{2} h_i f_t^2(x_i)\right] + \gamma T + \tfrac{1}{2}\lambda \sum_{j=1}^{T} w_j^2, \tag{2}$$

where $g_i = \partial l(y_i, z)/\partial z$ and $h_i = \partial^2 l(y_i, z)/\partial z^2$ are evaluated at $z_i^{(t-1)}$, and $f_t(x_i) = w_{q(x_i)}$ is the leaf weight that row i reaches, with $q(x_i)$ the leaf index. For the logistic loss, $g_i = \hat{p}_i - y_i$ and $h_i = \hat{p}_i(1 - \hat{p}_i)$.

For a fixed tree structure, the optimal weight of leaf j is

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{3}$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ are the sums over the rows I_j that reach leaf j. The gain of a split is

$$\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda}\right] - \gamma, \tag{4}$$

with G_L and H_L for the left child and G_R and H_R for the right child. A split is kept only when the gain is positive. The tree also stops at the maximum depth or when a leaf has fewer than the minimum child weight.

Depth, learning rate, tree count, and the row and column subsampling rates are tuned on the 14,000 training rows. Each seed runs its own 30-configuration search with stratified group five-fold cross-validation, and the grouping keeps both answers to the same question in one fold. The seed-42 values are in Section 4, and the per-seed values are in the repository. Each seed trains on the 14,000 rows with early stopping on the 3,000 validation rows (30 rounds). The 300-tree setting is the tuned upper bound, and seed 42 stops at 209 trees. The headline comparison uses the uncalibrated probability at a fixed 0.5 threshold on the 3,000 test rows. Platt and isotonic maps are fitted on the same validation rows for the calibration study. The source-only P0 threshold is selected on those rows at a 5% false-positive budget. The test rows are never used for tuning, early stopping, calibration, or threshold selection. The final step returns the mean over the three seeds. The search leaves γ, λ, and the minimum child weight at their XGBoost defaults. Algorithm 1 summarizes the tuning, training, and calibration steps.

Algorithm 1 Baseline feature-based XGBoost (training and evaluation)

Require: labeled rows (q_i, c_i, a_i, y_i) and seeds $\mathcal{S} = \{42, 123, 456\}$
Ensure: trained model, test metrics, and Platt and isotonic maps

```
extract the 26 base features for every row
for each seed s ∈ S do
    tune XGBoost on the 14,000 training rows with 30 random configurations and stratified group 5-fold cross-validation
    train the classifier on the same rows with early stopping on the 3,000 validation rows
    evaluate the raw score on the 3,000 test rows at the fixed 0.5 threshold
    fit Platt and isotonic maps on the validation rows for the calibration study
end for
return the mean metrics over the seeds
```

3. Proposed Method

3.1. Working Principle

The task is to estimate $P(\text{hallucinated} \mid q, c, a)$ for a question q, reference context c (if available), and a candidate answer a from a black-box LLM. The target is not truthfulness in general but evidence consistency: whether a follows from c or contradicts it, and the output is a probability that a decision threshold turns into a flag. Figure 1 shows the pipeline that computes this probability. The system takes the three inputs and extracts 35 features: 26 base features in seven groups, eight claim-level NLI aggregates, and a source indicator. EC-XGB maps the feature vector to a raw score, and a Platt calibrator fitted on the RAGTruth QA calibration rows maps the score to the displayed probability. The system reports the risk score, per-claim verdicts, and SHAP contributions. The lower band shows the deployment path: a Next.js front end calls a FastAPI service that loads the model, calibrator, explainer, and feature models at startup. The rightmost block lists the evaluation corpora. HaluEval measures in-domain performance, RAGTruth summarization and data-to-text measure adaptation to task types seen in training, and RAGTruth QA and FaithBench are held out.

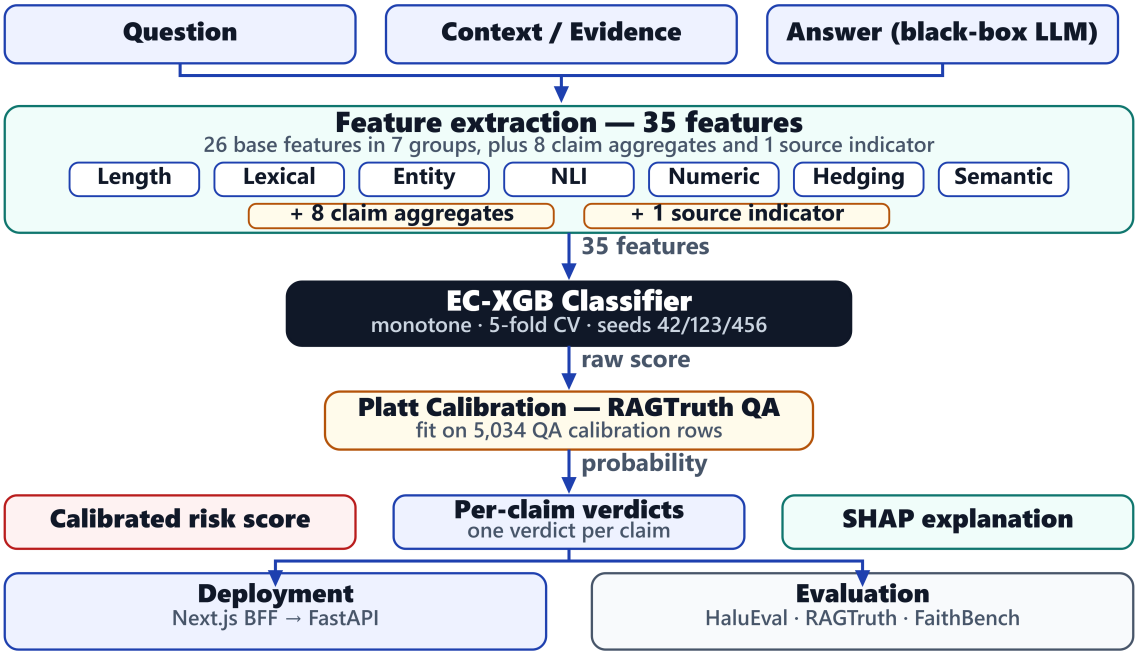


Figure 1: End-to-end pipeline of EC-XGB.

3.2. Features

The model uses 26 base features in seven groups (Table 1) and eight claim-level aggregates (Table 2). Natural language inference (NLI) probabilities come from a DeBERTa-v3 cross-encoder [20] trained on SNLI [21] and MultiNLI [22], semantic similarity uses MiniLM embeddings [23, 24], and entities come from spaCy named entity recognition (NER) [25]. Length is included because HaluEval labels longer answers as hallucinated more often, so it is a strong but fragile signal. Lexical overlap captures copying and is the strongest SHAP feature, but it cannot detect a fluent paraphrase that contradicts the evidence. A novel entity is a useful warning sign, although a correct answer can introduce a new name. The NLI features are the only group that represents logical support rather than surface overlap. A changed number is a common and easy-to-detect hallucination. Hedging words are weak on their own but separate a cautious correct answer from a confident wrong one. Semantic similarity catches paraphrase and topic drift that lexical overlap misses. The eight claim aggregates repeat the NLI signals at the level of atomic claims, which is the level a reviewer checks. RAGChecker also uses entailment-based claim metrics [26].

3.3. Evidence-Consistent XGBoost

EC-XGB changes the conventional setup in its feature treatment and its training rows. The character and word counts are log-scaled, and monotone constraints are placed on the evidence features. The constraints are negative on context overlap, context Jaccard, context entailment, context cosine, claim minimum and mean entailment, and the supported-claim ratio, and positive on context contradiction, novel-entity ratio, novel numbers, claim maximum and mean contradiction, and the contradicted-claim ratio. All other features are unconstrained. The eight claim features in Table 2 are added: seven NLI aggregates and a claim count. They use the same NLI scorer and support-first rule as the verification layer. A claim is supported when any retained sentence has an entailment probability of at least 0.5 and at least as high as its contradiction probability. It is contradicted when no sentence supports it and a sentence has a contradiction probability of at least 0.5 and above its entailment probability. Otherwise it is unsupported.

Consider an answer with K claims. Let $C(k)$ be the maximum contradiction probability of claim k over its selected context sentences, $E(k)$ the maximum entailment probability, and v_k its verdict. The aggregate vector is

$$\psi(x) = [\max_k C(k), \overline{C}, r_{\text{con}}, r_{\text{sup}}, r_{\text{unsup}}, \min_k E(k), \overline{E}, K], \tag{5}$$

Table 1
Base feature groups (26 features total).

Group	Features	Source
Length/style	n_chars, n_words, n_sentences, avg_word_len	regex
Lexical	overlap vs. context & question, Jaccard ×2	token sets
Entity	entity counts, overlap ratio, novel-entity ratio	spaCy NER
NLI	entail/contradict/neutral × 2 directions	cross-encoder NLI
Numeric	number counts, overlap ratio, novel numbers	regex
Hedging	hedge count, hedge density	lexicon
Semantic	cosine(context,answer), cosine(question,answer)	MiniLM

Table 2
Claim-level NLI aggregates (8 features).

Signal	Features	Source
Contradiction	max and mean contradiction, contradicted-claim ratio	NLI cross-encoder
Support	supported-claim ratio, unsupported-claim ratio, min and mean entailment	NLI cross-encoder
Decomposition	number of claims	sentence splitter

with

$$r_{\text{con}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{contradicted}], \quad r_{\text{sup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{supported}], \quad r_{\text{unsup}} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}[v_k = \text{unsupported}]. \tag{6}$$

Here $x = (q, c, a)$ is the input triplet, and $\overline{C}$ and $\overline{E}$ are the means of $C(k)$ and $E(k)$ over the claims. For $K > 0$, the three ratios sum to one. The model input has 35 dimensions: the 26 base features (two of them log-scaled), the eight claim features, and one source indicator.

The training set adds the RAGTruth summarization and data-to-text rows. The ablation ladder starts from m0, the 26-feature conventional reference, and adds log-scaled length counts with monotone constraints at m1, the eight claim features at m2 (34 features), and the RAGTruth non-QA rows with the source indicator at m3 (35 features). A binary source indicator is one for those rows and zero for HaluEval, so the classifier can separate the two sources. The classifier never sees RAGTruth QA rows, so its QA ranking is zero-shot. The display calibrator does see labeled QA rows below, so the display score is not zero-shot. The 35-feature combination is called EC-XGB, and seed 42 is the deployed checkpoint while the other seeds are a stability check. The ablation in Section 5 shows that the claim block adds no aggregate F1 gain, and the claim count K is kept because an answer with many claims is harder to verify. Algorithm 2 lists the procedure: steps 1 and 2 build the feature matrix, and steps 3 to 9 repeat the fit for the three seeds. Each repetition builds the 35-feature input, trains on HaluEval plus the RAGTruth non-QA rows, fits the display calibrator on the 5,034 QA calibration rows, and selects the source and target thresholds.

Algorithm 2 Evidence-consistent XGBoost (EC-XGB) training and deployment

Require: HaluEval QA training and validation rows, RAGTruth summarization and data-to-text training rows, RAGTruth QA calibration rows, seeds $\mathcal{S} = \{42, 123, 456\}$
Ensure: seed-42 model and display calibrator, source and target thresholds, per-seed metrics
extract the 26 base features and the eight claim features for every row
log-scale the character and word counts, and attach the monotone constraints
for each seed $s \in \mathcal{S}$ **do**
    build the 35-feature input, with the source indicator at zero for HaluEval and one for RAGTruth
    train XGBoost on the HaluEval rows plus the RAGTruth non-QA rows, and keep every RAGTruth QA row out of classifier training
    fit the Platt display calibrator on the RAGTruth QA calibration rows
    select the source threshold on the HaluEval validation split
    select the target threshold on the QA calibration rows at a 5% false-positive budget
end for
return the seed-42 model and calibrator, both thresholds, and the per-seed metrics

3.4. Calibration and Explainability

Source calibration uses the HaluEval validation split. Platt scaling fits a sigmoid $p(x) = \sigma(Af(x) + B)$ to the uncalibrated probability $f(x)$, where A and B are fit by maximum likelihood [27]. Isotonic regression fits a non-decreasing step function g so that $p(x) = g(f(x))$ [28]. Both are compared with the uncalibrated probability on the HaluEval test split. The comparison reports expected calibration error over 10 equal-width bins,

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{7}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the empirical positive rate and mean confidence in bin B_m, the Brier score $\mathrm{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$, and reliability diagrams [17]. Post-hoc calibration degrades when the target distribution moves [29].

The second protocol fits the calibrator on labeled target rows and evaluates it on a disjoint target test split. The deployed EC-XGB display score uses a Platt calibrator fitted on the 5,034 RAGTruth QA calibration rows. Those rows are scored with the source indicator set to one, and the API uses the same value for every request. This target calibration needs labeled target rows and is not zero-shot. The source-only protocol (P0 in Table 4) fits both the calibrator and the threshold on the HaluEval validation split. The deployed calibrator is Platt. An earlier deployment fit compared Platt and isotonic on the test split, and Section 5 records that choice.

A TreeExplainer computes exact SHAP values [18] for the raw XGBoost model. Global importance is the mean absolute SHAP value over a stratified test sample, and the explanation endpoint returns the five features with the largest absolute value for one prediction. Post-hoc attributions can be unstable even when the model is fixed [30], so the pilot evaluates them directly [31]. A 20-case pilot applies six operators: entity, number, and date replacement, support removal, context insertion, and clause shuffling. It reports the mean absolute score change, the fraction of large changes, the top-1 flip rate, and the Spearman rank correlation between the original and perturbed rankings.

3.5. System and Reproducibility

The method is deployed in a web application with a Next.js front end and a FastAPI backend. The API loads the EC-XGB model, the Platt calibrator, the SHAP explainer, and the NLI, spaCy, and MiniLM feature models at startup. The deployed checkpoint is the 35-feature m3 model with seed 42. Section 5 compares it with B2, the conventional 26-feature checkpoint tuned on HaluEval QA, and with B6 m0, the fixed-hyperparameter 26-feature reference from the same tuning protocol that lacks the EC-XGB changes. Claims are split from the answer and checked with sentence-level NLI against the provided context, a local passage index, or a web search API. A claim is supported when any retained sentence has an entailment probability of at least 0.5 and at least as high as its contradiction probability. A contradiction decides only when no sentence supports the claim. A relevance gate drops sentences with no claim vocabulary, and the verdicts are shown per claim.

Figure 2 shows the deployment view. The browser calls the Next.js backend-for-frontend (BFF), which streams chat tokens and forwards tool calls to the model. The analyze route calls FastAPI through the `/api/ml/*` rewrite. FastAPI

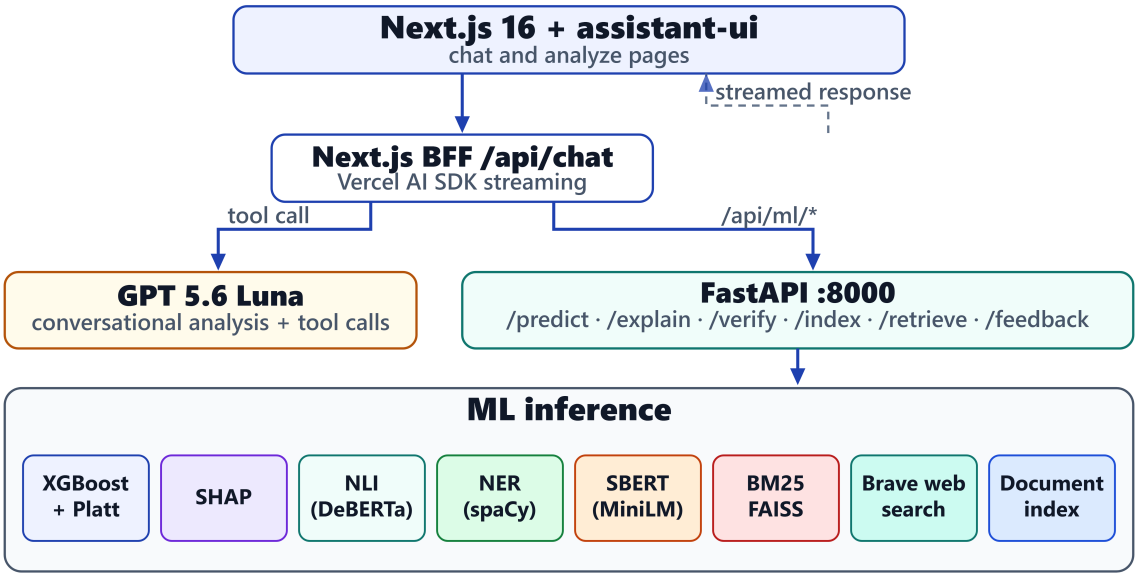


Figure 2: Deployment architecture of the HaluRISC system.

loads EC-XGB, the Platt calibrator, SHAP, NLI, spaCy, and MiniLM at startup. The lower band lists the inference components. A tool call goes to the judge, and a prediction request goes to `/predict`, `/explain`, or `/verify`. Figure 1 shows how one answer is scored, and Figure 2 shows where each component runs. The card's headline follows the claim verdicts when they are present. A contradicted claim sets high risk, an unsupported claim sets medium risk, and a fully supported answer sets low risk. The percentage is the display score: the chat card adjusts it by the claim mix, and the analyze card reports the unadjusted Platt score.

In the hallucinated-number case, the context says 5 days per decade and the answer says 10. The raw score is 0.999 and the target-calibrated score is 35.4%, which the card rounds to 35% and places in the medium band (0.35–0.60). The claim view marks the number claim unsupported, so the chat card also reports medium risk. The two labels agree here, but they come from different rules. The claim-verdict audit is reported in Section 5. The requirements file pins every dependency, split indices are stored with the artifacts, and the audit scripts check the rebuilt and unchanged reported values against the frozen artifacts. The audits run after every build and fail on any mismatch.

The contribution is the training and evaluation protocol, not a new learner. EC-XGB combines three standard changes: log-scaled length counts with monotone constraints, claim-level NLI aggregates for explanation, and multi-source training with a source indicator. Section 5 measures what each change contributes. The claim aggregates are kept for per-claim explanation and do not add an aggregate F1 gain. The seen-task gain comes from the m3 stack that adds the RAGTruth rows and the source indicator, and the two are not separated. The risk score needs no LLM judge calls. The optional claim judge is a separate verification step, and the reported EC-XGB results do not use it.

4. Experiments

4.1. Experimental Setup

All experiments were run on a Windows laptop with an NVIDIA RTX 3060 6 GB GPU, CUDA 12.8, torch 2.11.0+cu128, and 32 GB of RAM. The code runs on Python 3.12 with scikit-learn 1.9.0, XGBoost 3.3.0, SHAP 0.52.0, spaCy 3.8.14, and sentence-transformers 5.6.1. Each run can be reproduced because of saved split indices and centralized seeds. A randomized search was performed over 30 hyperparameter configurations with grouped five-fold cross-validation. For the seed-42 conventional model, the configuration has a maximum tree depth of 5, a learning rate of 0.05, 300 trees, a subsampling ratio of 0.7, and a column sampling ratio of 0.7. Each seed is tuned separately for the main comparison, and the component ablation in Section 5 holds one configuration fixed across variants so that the feature effects are not confounded with tuning. Seeds 42, 123, and 456 are repeated where a seed applies; the single-seed tables are labeled.

4.2. Dataset

HaluEval QA [32] provides 10,000 question pairs, each with a knowledge passage, a correct answer, and a hallucinated answer. The split is 70/15/15 by item, so both answers to a question stay together. HaluEval is entirely ChatGPT-generated [32, 33], so the boundary between a factual error and a hallucination is not always clean. RAGTruth [34] supplies natural responses with human span annotations, and FaithBench [13] supplies modern-LLM summarization hallucinations. RAGTruth keeps its official document-level split: summarization and data-to-text supply the training rows and a held-out test set each, and QA supplies 5,034 calibration rows and a held-out test set. RAGTruth QA is a subset of the full corpus, so its rows are not additive. The three held-out test splits form the pooled RAGTruth test set. Table 3 lists the corpora, sizes, and positive rates. MedHallu pqa_labeled [15] adds medical QA pairs with human-annotated gold answers and paired LLM-written hallucinations. HaluBench [16] contributes a stratified 30% sample of its DROP, FinanceBench, and covidQA rows, balanced across PASS and FAIL; its HaluEval, RAGTruth, and PubMedQA rows are excluded. A seeded sample from each HaluEval task forms the task-shift sets. These external corpora are never used for fitting or threshold selection.

Table 3
Datasets used for training and evaluation.

Corpus	Rows	Groups	Positive	Use
HaluEval QA	20,000	10,000	50.0%	train / validation / test
RAGTruth QA	5,934	989	17.8 / 31.1%	held-out test + calibration
RAGTruth full	17,790	2,965	34.9%	train + pooled test (2,700)
FaithBench	750	750	68.1%	zero-shot
MedHallu	2,000	1,000 pairs	50.0%	zero-shot external
HaluBench subset	900	3 sources	50.0%	zero-shot external
HaluEval dialogue	2,000	1,000 pairs	50.0%	task-shift test
HaluEval summarization	2,000	1,000 pairs	50.0%	task-shift test

Leading and trailing whitespace is trimmed, and lexical matching is case-folded. NER and NLI read the original text, and entity strings are lowercased for matching. A missing context is a special case: lexical overlap is 0 and NLI probabilities are uniform. In m1 and later variants, the two length counts are log1p-transformed; m0 and the conventional model keep raw counts. Claims are split into at most six clause-level segments, with conjunctions broken into separate clauses. When the context has more than four sentences, each claim is matched to its four best sentences by token overlap, and each side of a pair is truncated to 400 characters. Both index sets are frozen. A RAGTruth answer is positive when it carries at least one annotated hallucination span. FaithBench severity annotations are reduced by a worst-severity rule, and a Questionable annotation counts as positive. HaluEval uses its native label. NLI and SBERT inference ran on the reference GPU in fp16, while lexical and entity features run on the CPU. The feature-based logistic baselines standardize their inputs on the training split; the TF-IDF models use sparse features, and XGBoost does not scale. The source indicator is 1 for FaithBench and MedHallu and 0 for HaluBench and the task-shift sets. FaithBench and RAGTruth summarization and data-to-text have no question, so question overlap is 0 and question similarity is the cosine between the answer and the encoded empty string. Licenses and the data-availability statement are in the Declarations.

4.3. Baselines and Experimental Procedure

Twelve models are tested. The heuristic sets risk to one minus lexical overlap, and a length-only logistic control uses the two log-scaled length counts. Logistic regression and a 300-tree random forest are the classical comparators. The conventional XGBoost model [19] uses the 26 base features. Three further controls isolate the feature design: NLI-only, answer-only TF-IDF, and full-input TF-IDF. A majority classifier, a fine-tuned RoBERTa-base [35], and a reduced reimplementation of the lightweight ensemble of Sundaragiri et al. [10] complete the set; the two judges are in Section 5. Internal-state detectors are outside the black-box setting [36]. The RoBERTa baseline receives up to 256 tokens and trains for three epochs. The majority baseline predicts the negative class, so probability metrics are undefined. The external ensemble uses five features (ROUGE-L, SBERT cosine, two NLI probabilities, answer length) in a logistic regression fit on the 14,000 training rows, and a validation-selected threshold of 0.55 gives F1 0.9514, so the fixed 0.5 threshold does not change the comparison. EC-XGB's display calibrator is fitted on the RAGTruth QA split; the in-domain Platt calibrator is fitted on validation. Metrics are reported to four decimals and rates to one.

Significance uses McNemar on seed-42 predictions, 2,000 two-level bootstrap resamples, and Wilcoxon across seeds. No question appears in both training and evaluation. Calibration is compared on raw, Platt, and isotonic scores; Table 4 defines P0, P1, and P2, and every protocol uses the same test rows.

Table 4
The three evaluation protocols.

Protocol	Calibrator fit	Threshold fit	Zero-shot
P0 source-only	HaluEval validation (3,000)	HaluEval validation	yes
P1 target-calibrated	RAGTruth QA calibration (5,034)	fixed at 0.5	no
P2 target-thresholded	RAGTruth QA calibration (5,034)	same rows at 5% FPR	no

The deployed system uses P1, and P2 is the thresholded alternative. Only P0 is zero-shot; P1 and P2 read labels from the target calibration split, which is disjoint from the test split. SHAP [18] TreeExplainers give global and local explanations. Two checks test reliability: the rank agreement between the m3 and B2 SHAP importances and a six-operator perturbation pilot. Feature latency is measured on 200 HaluEval rows on the reference laptop. The 900 RAGTruth QA rows are scored by two judges, GPT-5.6 Luna and Qwen 3.5 4B, under the LLM-as-a-judge protocol [37], and costs are computed from the recorded token counts.

4.4. Evaluation Metrics

Two families of metrics are reported. Threshold-dependent metrics measure the decision at a fixed cut-off, and threshold-free metrics measure how well a model ranks answers. Both are needed because a model can rank well and still make poor decisions when the threshold is fixed. Precision is the share of flagged answers that are truly hallucinated. Recall is the share of hallucinated answers that the detector catches, and F1 is their harmonic mean. The false-positive rate (FPR) is the share of correct answers that are flagged, which is the operating constraint under distribution shift. TP and TN are true positives and true negatives, while FP and FN are false positives and false negatives: a correct answer flagged as hallucinated, and a hallucinated answer that is not flagged. The flag rate is the share of answers above the decision threshold, and a rate near 100% means every answer reaches a human reviewer.

$$\text{Precision } (P) = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall } (R) = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad F_1 = \frac{2PR}{P + R}, \tag{8}$$

$$\text{False-positive rate (FPR)} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \qquad \text{Flag rate} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\hat{p}_i \geq \tau]. \tag{9}$$

The second group measures ranking, calibration, and paired error patterns. AUROC is the probability that a randomly chosen hallucinated answer receives a higher score than a randomly chosen correct answer, with 0.5 at chance. The area under the precision-recall curve (PR-AUC) is more sensitive than AUROC when the positive class is rare. RAGTruth QA has a positive rate of 17.8%, so PR-AUC is reported beside AUROC. The Matthews correlation coefficient (MCC) uses all four cells of the confusion matrix and ranges from -1 to 1, with zero at chance. The Wilson interval gives a 95% range for a proportion and is used for the P2 recall and false-positive rate, where the counts are small.

$$\text{AUROC} = \int_0^1 \text{TPR}(u)\, d\,\text{FPR}(u) = P(s^+ > s^-), \tag{10}$$

$$\text{PR-AUC} = \int_0^1 P(r)\, dr, \qquad \text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}, \tag{11}$$

$$\text{CI}_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}. \tag{12}$$

5. Results

5.1. Model Comparison

Table 5 reports the HaluEval test results for eleven baselines and EC-XGB, averaged over three seeds where a seed applies. EC-XGB leads the conventional model by 0.0009 F1, which is smaller than the width of the conventional seed-42 bootstrap interval [0.9812, 0.9895], so the benchmark does not resolve the ordering. The ranking columns repeat that pattern: EC-XGB is first among the feature-based models on AUROC and PR-AUC, the conventional model and random forest are close behind, and RoBERTa is slightly higher. Calibration gives a different summary, because the conventional model has the lowest ECE while EC-XGB and RoBERTa have lower Brier scores. The weak controls are NLI-only and full-input TF-IDF, while answer-only TF-IDF is much stronger, so adding the question and context hurts. The reduced external ensemble is below EC-XGB by 0.0337 F1 [0.0262, 0.0419] and 0.0135 AUROC [0.0098, 0.0177], but it is trained on fewer rows and with a narrower feature set, so that gap is descriptive. Figure 3 shows the curves behind the table, and the curves use seed 42 while the table values are three-seed means.

The XGBoost margin over the random forest is 0.0004 F1, with McNemar $p = 0.502$; the difference is not significant. The margin over logistic regression is 0.0187 F1 ($p = 1.86 \times 10^{-10}$) and over the heuristic 0.0495 ($p = 2.28 \times 10^{-30}$). A source-only encoder wins in-domain (F1 0.9912 vs 0.9855) but is near chance on each RAGTruth task (AUROC 0.458 on summarization and 0.491 on data-to-text, against 0.736 and 0.770 for EC-XGB on those tasks); pooling gives 0.337 for the encoder and 0.582 for EC-XGB, because the encoder's pooled score mixes between-task score shifts with within-task ranking. An adapted encoder on the same multi-source rows reaches 0.694 on the pooled RAGTruth test set, above EC-XGB's 0.582, so the seen-task gain is not specific to the feature model. Neither model generalizes to unseen QA (0.490 vs 0.549), and FaithBench is worse for both (0.434 vs 0.594). Adaptation helps any model on seen tasks; the feature model keeps its relative ordering on unseen tasks.

Table 5
Test-set performance on HaluEval QA, mean over seeds 42/123/456 where a seed applies; deterministic rows are single fits (n = 3,000).

Model	P	R	F1	AUROC	PR-AUC	MCC	ECE	Brier
Heuristic (1−overlap)	0.9473	0.9233	0.9352	0.9132	0.8196	0.8723	0.3541	0.2672
Length-only (2 feats)	0.9454	0.9460	0.9457	0.9759	0.9722	0.8913	—	—
Logistic regression	0.9768	0.9553	0.9660	0.9932	0.9901	0.9329	0.0099	0.0253
Random forest	0.9897	0.9789	0.9842	0.9980	0.9983	0.9687	0.0143	0.0127
NLI-only	0.6479	0.6967	0.6714	0.7177	0.7531	0.3189	0.0678	0.2182
TF-IDF (answer-only)	0.9550	0.8920	0.9224	0.9696	0.9665	0.8519	0.0831	0.0662
TF-IDF (full input)	0.6397	0.5647	0.5999	0.6754	0.6785	0.2484	0.0679	0.2328
External ensemble†	0.9508	0.9527	0.9517	0.9850	0.9856	0.9033	0.0141	0.0391
XGBoost (conventional)	0.9932	0.9762	0.9846	0.9979	0.9983	0.9697	0.0045	0.0117
EC-XGB (m3)	0.9939	0.9773	0.9855	0.9984	0.9987	0.9715	0.0074	0.0108
RoBERTa-base	0.9955	0.9869	0.9912	0.9990	0.9993	0.9825	0.0070	0.0081
Majority (negative)	0.0000	0.0000	0.0000	—	—	0.0000	—	—

5.2. EC-XGB and Ablation

Table 6 reports the component ablation under one fixed-hyperparameter configuration (max depth 5, learning rate 0.05, 300 trees, subsample 0.7, colsample 0.7, selected once on the m3 seed-42 run), so the columns measure feature effects rather than tuning effects. The ladder from m0 to m3 is defined in Section 3, and ECE is computed on the raw score. The in-domain F1 span is only 0.0006 across m0–m3. The claim aggregates change F1 by −0.0004 at m2 relative to m1, and the full m3 stack changes it by +0.0001; neither is an aggregate gain. On the pooled RAGTruth split, AUROC rises from 0.494 to 0.590 and raw ECE falls from 0.635 to 0.275; on held-out QA the comparison is 0.556 vs 0.558, within seed noise. McNemar gives $p = 0.789$ for m0 vs m3, so the ablation does not resolve a difference. The gain is not from the claim features; it appears when the multi-source rows and source indicator are added. The MedHallu source-zero check suggests the indicator is not the main driver, but a seen-task source ablation would settle it. The deployed checkpoint reaches 0.9855 under its own tuned configuration, 0.0001 above the fixed ablation.

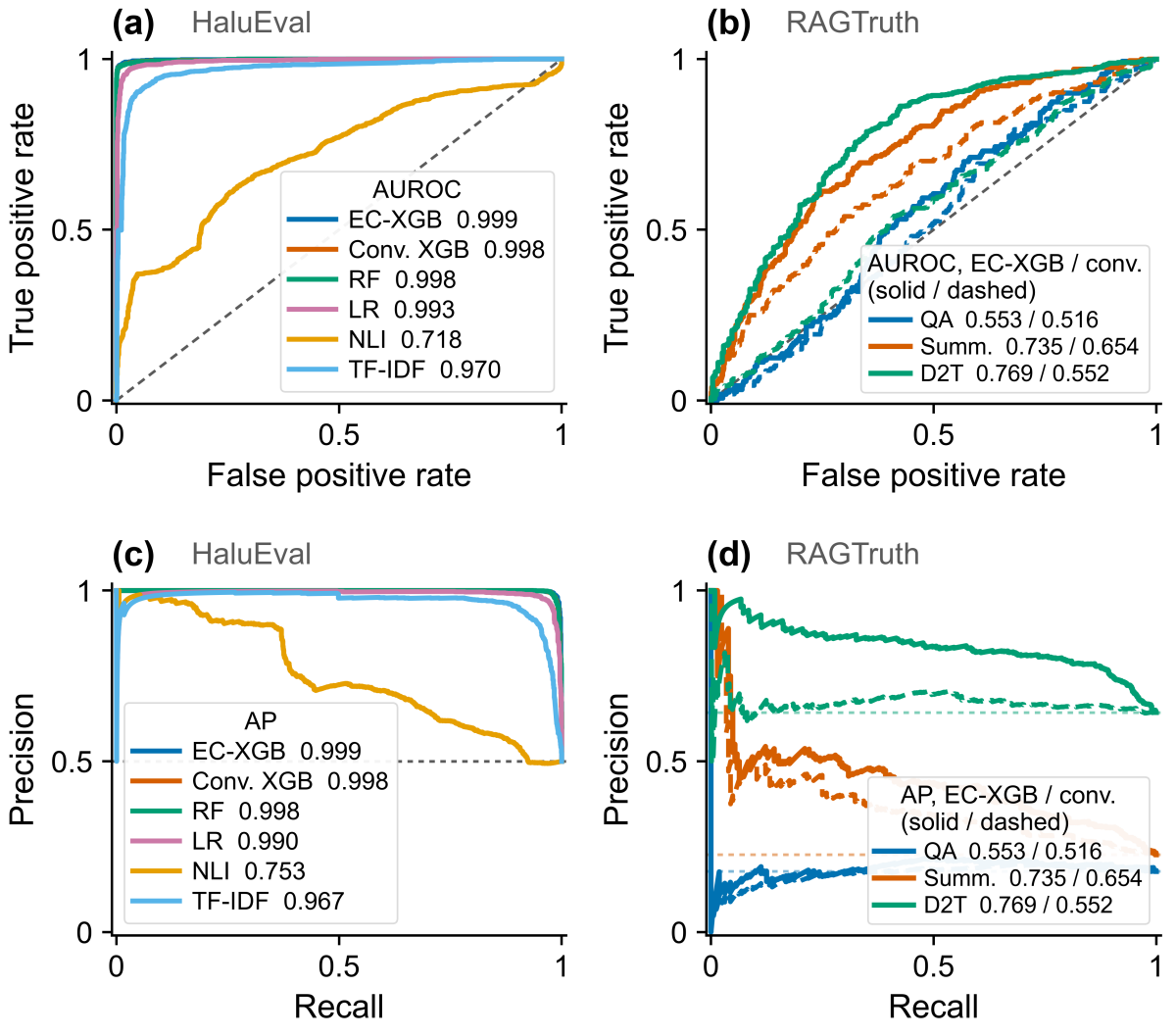


Figure 3: ROC and precision-recall curves on HaluEval and RAGTruth (seed 42).

Table 6
EC-XGB component ablation on the HaluEval test set with one fixed configuration, mean over three seeds.

Variant	Features	F1	AUROC	MCC	ECE	RAGTruth AUROC	RAGTruth ECE
m0 conventional	26	0.9849	0.9979	0.9701	0.0050	0.4937	0.6352
m1 debias	26	0.9853	0.9976	0.9710	0.0042	0.5567	0.6260
m2 + claims	34	0.9849	0.9985	0.9701	0.0051	0.5461	0.6359
EC-XGB (m3)	35	0.9854	0.9986	0.9713	0.0078	0.5898	0.2746

Table 7
Feature-group ablation of the fixed-hp m3 model on the HaluEval test set, mean over seeds 42/123/456.

Removed group	Features	ΔF1	ΔAUROC	ΔMCC
length	4	−0.0066	−0.0012	−0.0127
lexical	4	−0.0171	−0.0040	−0.0343
entity	4	−0.0010	+0.0000	−0.0020
nli	6	−0.0040	−0.0004	−0.0080
numeric	4	−0.0003	+0.0001	−0.0007
hedging	2	+0.0000	+0.0001	0.0000
semantic	2	0.0000	+0.0002	+0.0000
claims	8	−0.0010	−0.0003	−0.0020
source	1	−0.0001	+0.0001	−0.0002

Table 7 repeats the ablation on the fixed-hp m3 model, whose full reference reaches F1 0.9861, AUROC 0.9985, and MCC 0.9726. Lexical overlap is the largest contributor, followed by length and NLI. The claim block costs about one seed standard deviation, so an aggregate claim gain is not resolved. Entity, numeric, hedging, semantic, and source each change F1 by less than 0.0011. Lexical overlap and length therefore carry the in-domain score, and the remaining groups add ranking information without moving F1 far. Deltas are relative to the full m3 refit under the same configuration, trained on the same 24,056 rows and evaluated once on 3,000 test rows without tuning or early stopping. The refit reference differs from Table 6 by 0.0007 F1, within its seed standard deviation of 0.0010.

Figure 4 shows the seed-level spread behind Table 6. The F1 and AUROC panels are tight, with within-variant differences below 0.0007 F1 and 0.0005 AUROC. The ECE panel is noisier: m0 is stable across seeds, while m2 and m3 vary more. The mean ordering is m3 > m1 > m2 > m0, but the per-seed ranks flip and the differences are below seed noise. The ECE sign is stable: m0 is lower than m3 on every seed, so the ablation does not establish a calibration advantage for m3 in this sample.

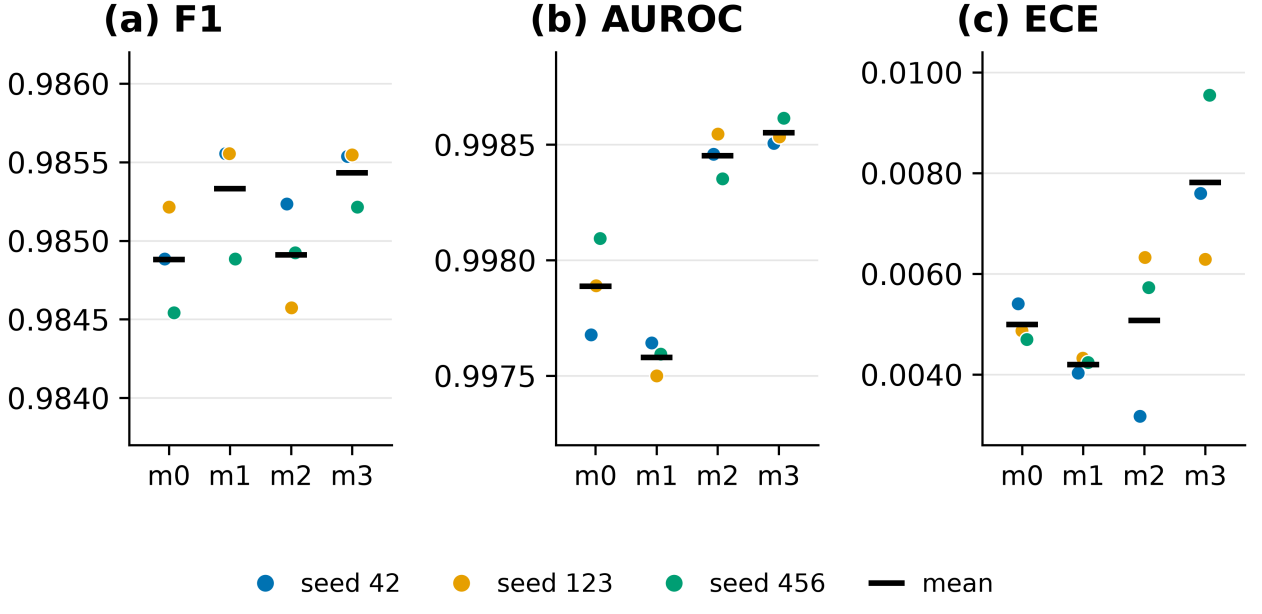


Figure 4: Per-seed F1, AUROC and ECE for the fixed-hyperparameter B6 variants.

5.3. Calibration, Shift, and External Validity

The raw, Platt-scaled, and isotonic versions of the conventional model give ECE 0.0045, 0.0091, and 0.0070 on HaluEval, so the calibrated ECEs are not lower in this sample. Figure 5 shows why. On HaluEval, the points at the two

ends sit on the diagonal; the middle of the score range holds few distinct values, so the curve has only a few stable bins. FaithBench is aggregated because its scores concentrate near 1.0. The Platt and isotonic curves overlap the raw curve at both ends. The m3 raw score is also well calibrated in-domain, with ECE 0.0078, so there is little room for calibration to improve on HaluEval. Platt scaling fits a two-parameter logistic map from the uncalibrated probability s, called the raw score in the figures, to a probability,

$$\hat{p} = \sigma(as + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}, \tag{13}$$

where a is the calibration slope and b is the intercept. A slope below one means the raw score is overconfident, and a slope near one means the scale is already usable. The Brier skill score (BSS) compares the calibrated score with a constant predictor at the evaluation prevalence,

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{prior}}}. \tag{14}$$

A positive BSS means the score beats the prior, and a non-positive BSS means it does not.

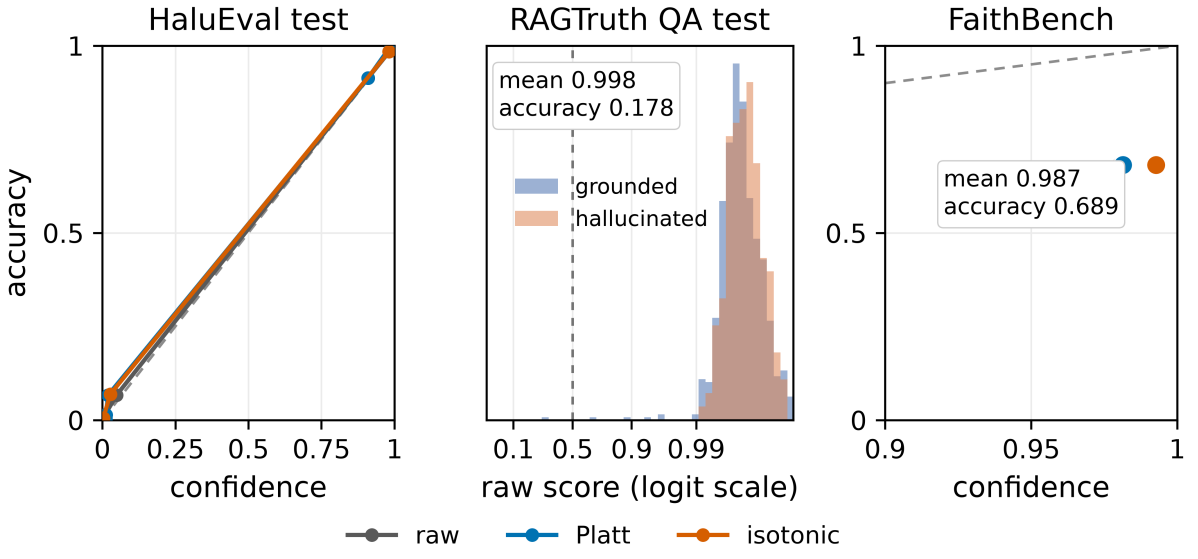


Figure 5: Reliability diagrams for the B2 model on HaluEval and FaithBench, and the RAGTruth QA raw-score distribution.

Source Platt cannot repair the base-rate shift. The fixed-hp m3 raw score is severely overconfident on RAGTruth QA: ECE 0.731 and Brier 0.689, with a Cox slope of 0.18 on the 5,034 calibration rows. Target Platt lowers ECE to 0.125 and Brier to 0.161, but the Brier skill score against the test prior is −0.10, and a constant predictor at the test prevalence has Brier 0.146, lower than every calibrated row. Calibration fixes the scale, not the ranking; AUROC stays at 0.558. The conventional target-Platt row behaves the same way at ECE 0.134 and Brier 0.164. Figure 6 shows the raw and target-recalibrated scores. The source-only operating point (P0) selects the threshold on the HaluEval validation split at a 5% false-positive budget. On HaluEval test, recall is 99.0–99.5% and FPR stays near 5%, but on RAGTruth QA the same threshold flags all 900 rows of the deployed seed-42 checkpoint, because every score exceeds 0.095, so P0 is not usable. Table 8 reports both models on the same test set. A prevalence-only predictor at 0.1778 reaches ECE 0.000 and Brier 0.146 and predicts no positives; at the calibration prevalence of 0.3107 it gives ECE 0.133 and Brier 0.164, so the calibrator sets the level to the calibration prevalence and the residual ECE is dominated by the prevalence gap. The conventional target-thresholded mode (P2) selects 0.313 at a 5% false-positive budget and reaches 5.0% recall at 5.8% FPR (precision 15.7%, F1 0.076). At 17.8% prevalence, random flagging at 5.8% FPR yields 5.8% recall, so P2 is chance-level. At 0.5, P1 flags no RAGTruth QA rows; for the deployed seed-42 calibrator, the maximum target-Platt score is 0.355 (35.4% before rounding, 35% on the card), the median is 0.331, and 16% of QA rows reach 0.35, so the card usually relies on the per-claim verdict adjustment. Figure 7 shows the operating point and label budget: mean

ECE is 0.130 at 50 rows, 0.124 at 252 rows, and 0.125 at 5,034 rows, but a single 252-row draw can miss the full-split ECE by about 0.06. All calibration metrics refer to the pre-verdict score; the displayed percentage is a design aid, not a validated probability.

Table 8
Calibration of the fixed-hp m3 model and the conventional target-Platt reference on the RAGTruth QA test set (n = 900, 17.8% positive). *Slope: three-seed mean; bracket: envelope of the three per-seed 95% intervals, not a pooled interval.

Predictor	Fit	ECE	Brier	BSS	Slope	Flagged
Prior (test prevalence)	constant 0.1778	0.000	0.146	0.000	—	0.0%
Prior (cal prevalence)	constant 0.3107	0.133	0.164	−0.12	—	0.0%
m3 raw	none	0.731	0.689	−3.71	0.18 [0.12, 0.27]*	96.8%
m3 source Platt	HaluEval val	0.765	0.742	−4.07	—	96.8%
m3 target Platt	RAGTruth cal	0.125	0.161	−0.10	—	0.0%
Conventional target Platt	RAGTruth cal	0.134	0.164	−0.12	—	0.0%

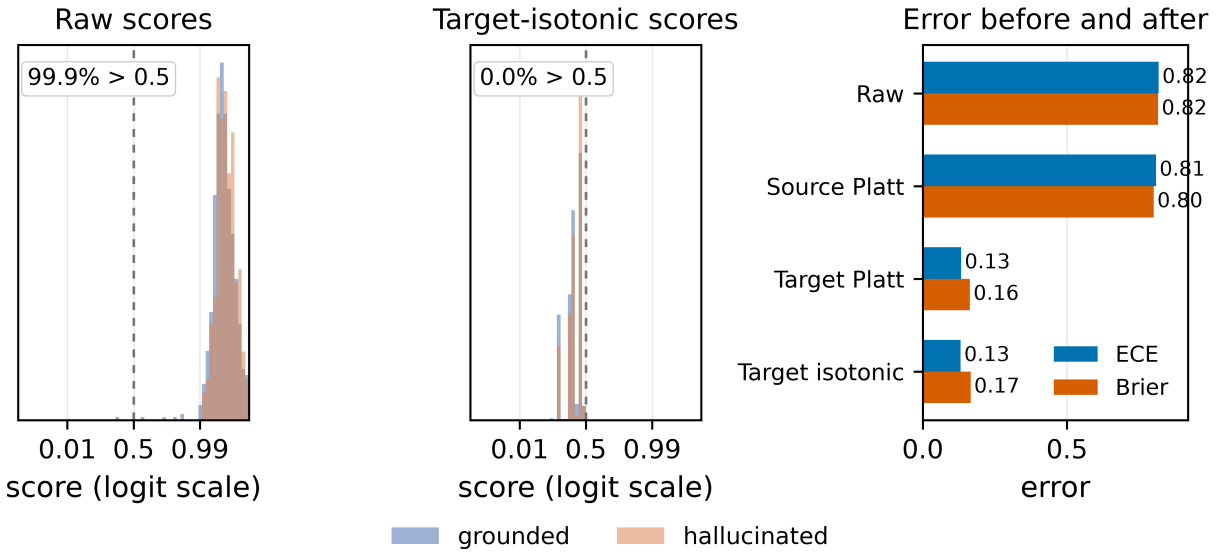


Figure 6: Target-domain recalibration of the conventional model on RAGTruth QA.

Table 9 reports the coverage. Split conformal prediction adds a coverage guarantee to the score. The nonconformity score for label y is $s(x, y) = 1 - \hat{p}_y(x)$, where $\hat{p}_1 = \hat{p}$ and $\hat{p}_0 = 1 - \hat{p}$. The calibration quantile and the prediction set are

$$\hat{q} = s_{(k)}, \quad k = \lceil (n+1)(1-\alpha) \rceil, \qquad C(x) = \{y : s(x, y) \le \hat{q}\}, \tag{15}$$

where $s_{(1)} \le \cdots \le s_{(n)}$ are the ordered calibration scores; this finite-sample order statistic yields coverage of at least $1 - \alpha$ for the prediction set. A class-conditional version replaces the single quantile with one quantile per label,

$$\hat{q}_y = s_{y,(k_y)}, \quad k_y = \left\lceil (n_y+1)(1-\alpha) \right\rceil, \qquad C(x) = \{y : s(x, y) \le \hat{q}_y\}, \tag{16}$$

where $s_{y,(1)} \le \cdots \le s_{y,(n_y)}$ are the ordered scores in class y and n_y is the number of calibration rows with label y. The main run fits the Platt map and the quantile on the same 5,034 rows, so its coverage is a construction check; it gives 0.940 and 0.967 coverage on QA and 0.900 and 0.950 on the calibration rows. A disjoint sensitivity fits Platt on 2,514 rows and the quantile on 2,520 rows. On that split, the 900 QA test rows reach 0.944 coverage at 90%, so the marginal guarantee holds. The marginal guarantee hides the class split: positive coverage is 0.685, while negative

coverage is 1.000. A class-conditional quantile on the full calibration rows raises positive coverage to 0.887, still slightly below nominal. FaithBench coverage falls to 0.320 and positive coverage to 0.003, because a Platt map fitted on QA sends almost every FaithBench score to the negative side. Conformal prediction therefore certifies the marginal rate only when the calibrator and the quantile are fitted on disjoint rows, and it does not transfer across the task shift. Multicalibration by answer length and claim count shows where the average ECE hides error. On the 900 QA test rows, the target-Platt score has overall ECE 0.125. The worst length bin is 60 words or fewer (n=154), with ECE 0.275; the worst claim-count bin is a single claim (n=36), with ECE 0.346. The Brier score does not follow the ECE ranking, so the average is not a safe summary for short or one-claim answers. FaithBench is not improved by the QA calibrator: its target-Platt ECE is 0.629 against 0.453 for the raw score.

Table 9
Split and class-conditional conformal coverage on RAGTruth QA and FaithBench, mean over seeds 42/123/456. Class-conditional rows use the full 5,034 calibration rows; split rows use the disjoint halves.

Dataset	Method	α	Coverage	Pos.	Neg.	Set size
QA test	split	0.10	0.944	0.685	1.000	1.59
QA test	split	0.05	0.972	0.842	1.000	1.75
QA test	class-cond.	0.10	0.886	0.887	0.886	1.69
QA test	class-cond.	0.05	0.936	0.925	0.938	1.80
FaithBench	split	0.10	0.320	0.003	1.000	1.00
FaithBench	split	0.05	0.323	0.006	1.000	1.01

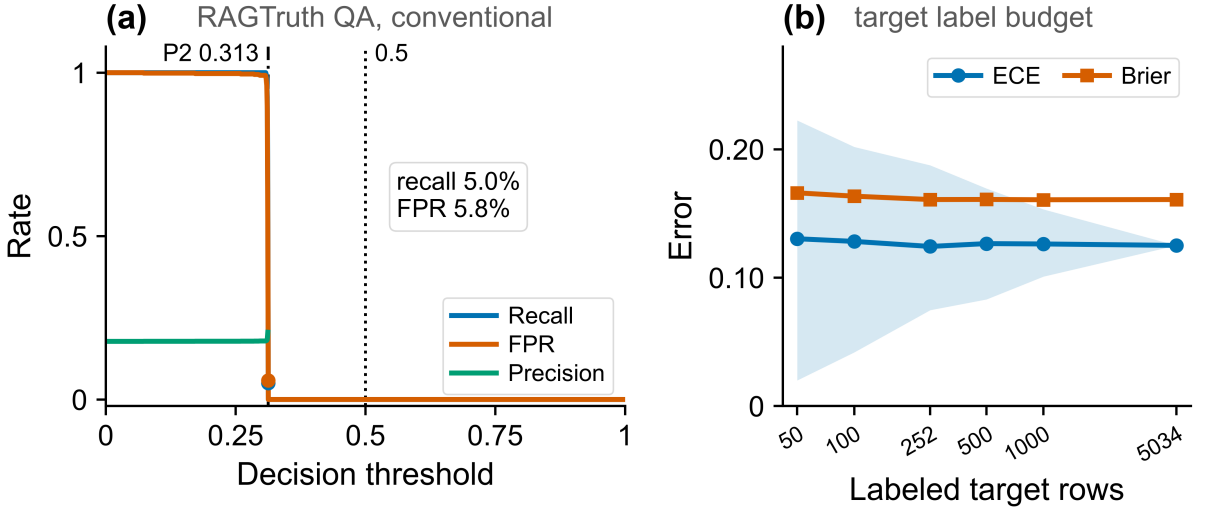


Figure 7: Operating point for the conventional target-Platt model and 100-draw label budget for the fixed-hp m3 calibrator on RAGTruth QA.

The conventional reference m0 is trained only on HaluEval, whereas EC-XGB also sees RAGTruth summarization and data-to-text rows, so those two pairs measure adaptation, not general shift (Table 10). On the pooled RAGTruth test split, the flag rate falls from 99.9% to 61.3% and AUROC rises from 0.475 to 0.582. The gain is task-specific: summarization AUROC rises from 0.634 to 0.736 and data-to-text from 0.567 to 0.771. Held-out QA remains near chance (AUROC 0.547 to 0.549). FaithBench AUROC rises from 0.509 to 0.594, but F1 falls from 0.813 to 0.120 and raw ECE rises from 0.292 to 0.450, so the raw scores transfer poorly. The conventional F1 matches the all-positive baseline because it flags almost every row. ECE uses raw scores, F1 uses a fixed 0.5 threshold, and Flagged is the share above it. Tables 7, 8, 11, and 17 report the fixed-hp refit; Table 10 reports the tuned B6 checkpoint.

The task-shift test is the weakest. On dialogue, m3 flags 99.3% of rows and its F1 equals the all-positive baseline, with AUROC 0.763; the length-only baseline reaches 0.724. On summarization, m3 reaches AUROC 0.717 and flags

Table 10
External and adaptation performance of B6 m0 and EC-XGB m3 from the tuned B6 run, mean over three seeds.

Dataset / task	Model	F1	AUROC	PR-AUC	Flagged	ECE
RAGTruth pooled	Conventional (m0)	0.5181	0.4753	0.3139	99.9%	0.6346
RAGTruth pooled	EC-XGB (m3)	0.5608	0.5815	0.3588	61.3%	0.2782
RAGTruth QA	Conventional (m0)	0.3022	0.5465	0.1911	99.9%	0.8149
RAGTruth QA	EC-XGB (m3)	0.3081	0.5491	0.1884	97.4%	0.7433
RAGTruth summarization	Conventional (m0)	0.3702	0.6343	0.3536	99.8%	0.7499
RAGTruth summarization	EC-XGB (m3)	0.3434	0.7358	0.4449	11.6%	0.0519
RAGTruth data-to-text	Conventional (m0)	0.7830	0.5665	0.6831	100.0%	0.3390
RAGTruth data-to-text	EC-XGB (m3)	0.8234	0.7705	0.8374	74.9%	0.0527
FaithBench	Conventional (m0)	0.8130	0.5089	0.6735	99.5%	0.2917
FaithBench	EC-XGB (m3)	0.1201	0.5941	0.7109	7.3%	0.4500

37.5%, while the length-only baseline reaches 0.788 and flags every row. The conventional m0 model flags 99.2% of dialogue rows and 97.7% of summarization rows, with AUROC 0.741 and 0.605. On dialogue, both models collapse to the all-positive baseline in F1, though their AUROCs stay above chance; on summarization, both rank below the length-only model. Figure 8 shows the raw score distributions of the conventional model: the HaluEval panel separates the classes, while the external panels pile both classes near 1.0, so the 0.5 threshold flags almost every row. This is the visual form of the 99.9% flag rate in Table 10.

Table 11
External validity on corpora never used for training, calibration, threshold selection, or model choice.

Dataset (n)	Model	F1	AUROC	ECE	Flagged	Pair acc.
MedHallu (2,000)	EC-XGB (m3)	0.667	0.821	0.493	100.0%	0.867
	Conventional (m0)	0.667	0.781	0.499	100.0%	0.822
	Length-only	0.667	0.260	0.500	100.0%	0.200
HaluBench (900)	EC-XGB (m3)	0.588	0.600	0.334	50.3%	—
	Conventional (m0)	0.585	0.609	0.346	51.6%	—
	Length-only	0.441	0.514	0.420	33.2%	—
HaluEval dialogue (2,000)	EC-XGB (m3)	0.670	0.763	0.488	99.3%	—
	Length-only	0.680	0.724	0.438	95.9%	—
HaluEval summarization (2,000)	EC-XGB (m3)	0.583	0.717	0.113	37.5%	—
	Length-only	0.667	0.788	0.500	100.0%	—

Table 11 reports the three external corpora and the task-shift pair. MedHallu is the strongest evidence that the ranking survives a domain change and the clearest evidence that the probability does not. The fixed-hp m3 model ranks the paired hallucinated answer above the grounded answer in 86.7% of 1,000 medical pairs [84.5, 88.7], with AUROC 0.821 [0.806, 0.837] on the 2,000 rows. The pair accuracy over M pairs is

$$\mathrm{PA} = \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\left[\hat{p}(a_m^+) > \hat{p}(a_m^-)\right], \tag{17}$$

where a_m^+ is the hallucinated answer and a_m^- the grounded answer. The conventional m0 model reaches 82.2% and 0.781. The length-only baseline is inverted, at 20.0% and 0.260, because grounded MedHallu answers are longer than the paired hallucinations (39.7 vs 27.8 words), the opposite of HaluEval (2.2 vs 11.0). The ranking therefore transfers, but the scale does not: m3 flags every row, so its F1 is the all-positive 0.667 and its ECE is 0.493. The source indicator is one for MedHallu and zero for HaluBench; a source-zero sensitivity on MedHallu changes pair accuracy by at most

0.003 and AUROC by at most 0.002, so the indicator does not drive the result. HaluBench is a 30% stratified sample. Its overall AUROC is 0.600, with covidQA 0.833, DROP 0.513, and FinanceBench 0.527; the model transfers to the biomedical subset but is at chance on the other two. The length-only baseline is at 0.514 overall, so 0.600 is not a length artifact. All evaluation sets are balanced 50/50 by construction, so F1 and the flag rate are prevalence-dependent and do not describe natural deployment; HaluBench is unpaired, so pair accuracy is undefined, and the length-only baseline is fit once on HaluEval train only.

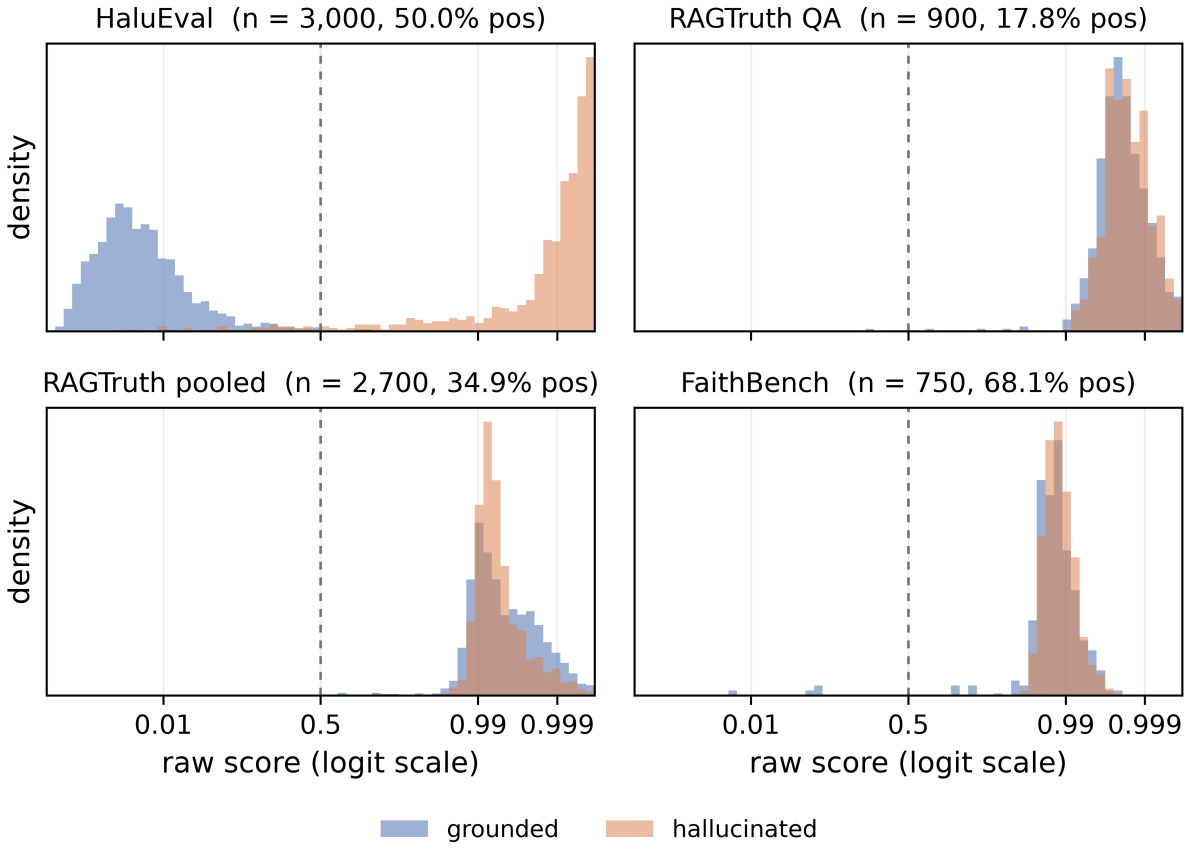


Figure 8: Raw score distributions of the conventional B2 model on HaluEval and the external test rows.

An index stress test asks whether the weak held-out QA result is a retrieval problem. The hybrid BM25 and FAISS index covers the 989 RAGTruth QA source documents. On a 300-row, 50-group sample, recall@1 is 0.94, recall@5 and recall@10 are 1.00, and MRR is 0.959. Replacing the gold context with the top-5 retrieved passages changes AUROC from 0.545 to 0.521, F1 from 0.345 to 0.349, and ECE from 0.706 to 0.704. The AUROC delta is −0.023 [−0.061, 0.011], which includes zero. Retrieval is therefore not the bottleneck: the model is near chance on RAGTruth QA even with the gold source passage. The qrels are the known source passage, not human web-search judgments, so this measures index recall, not end-to-end search quality. The sample has 50 source groups, so the interval is wide.

5.4. Statistical Analysis

Statistical tests use the paired fixed-hyperparameter predictions and 2,000 two-level bootstrap draws, where groups are resampled within each seed and then seeds are resampled. McNemar's test compares the discordant counts b and c on the paired seed-42 rows, where b is the number of rows where the first model is right and the second is wrong, and c is the reverse; the statistic uses a continuity correction. Holm's method controls the family-wise error rate across the four baseline comparisons. The 95% interval is the 2.5th and 97.5th percentile of the paired bootstrap deltas. The two-level version resamples groups within each seed before resampling seeds. The Wilcoxon signed-rank test compares the two models across the three seeds. Table 12 gives the McNemar counts. EC-XGB and the conventional model are

not significantly different under the deployed checkpoints ($p = 0.45$), and the fixed-hyperparameter ablation gives $p = 0.789$ on the same test rows. The fixed-hp m0-vs-m3 comparison is the primary test, and it is not part of the Holm family, which covers the four baseline comparisons against the conventional model. The test uses the single configuration of Table 6. The random forest matches the conventional model ($p = 0.502$). Logistic regression and the heuristic are significantly worse ($p = 1.86 \times 10^{-10}$ and $p = 2.28 \times 10^{-30}$), and both survive Holm correction. The Wilcoxon test across three seeds cannot reach significance (minimum $p = 0.25$). Table 13 reports the two-level bootstrap. On HaluEval, the m3−m0 F1 delta is +0.0006 [−0.0010, 0.0021], which includes zero, and the AUROC delta is +0.0007 [0.0002, 0.0012], which excludes zero by a small margin. The m2−m1 F1 delta is −0.0004 [−0.0020, 0.0012], and the m3−m1 delta is +0.0001 [−0.0015, 0.0018], so the claim features do not add an aggregate gain. Their AUROC deltas are +0.0009 and +0.0010, both with intervals above zero, so the added features reorder a few rows without changing F1.

On held-out RAGTruth QA the m3−m0 F1 delta is +0.0076 [0.0045, 0.0109], but the AUROC delta is +0.0017 [−0.0270, 0.0291], which includes zero. The F1 gain comes from flagging almost every row, not from better ranking. On the pooled seen tasks the F1 delta is +0.0451 [0.0348, 0.0554] and the AUROC delta is +0.0962 [0.0644, 0.1295], which is the adaptation gain. The paired fixed-minus-tuned contrasts all include zero, so the rebuild does not change these conclusions. HaluEval groups are item pairs (1,500), RAGTruth QA has 150 source groups, and the pooled split has 450; the QA AUROC interval includes zero, so the held-out transfer result is not resolved, and the pooled row measures adaptation to seen task types, not unseen-task transfer.

Table 12
McNemar tests on the paired seed-42 HaluEval predictions.

Comparison	b	c	p	Holm p
EC-XGB vs conventional	10	6	0.453	—
EC-XGB vs conventional (fixed hp)	8	6	0.789	—
Conventional vs random forest	12	8	0.502	0.502
Conventional vs logistic regression	69	11	1.86×10^{-10}	3.7×10^{-10}
Conventional vs heuristic	158	9	2.28×10^{-30}	6.8×10^{-30}
Conventional vs NLI-only	1006	26	5.6×10^{-204}	2.2×10^{-203}

Table 13
Two-level bootstrap deltas for the fixed-hyperparameter ablation (2,000 draws; groups within seed, then seeds). Intervals are 95%.

Dataset	Comparison	ΔF1 [95% CI]	ΔAUROC [95% CI]
HaluEval	m3−m0	+0.0006 [−0.0010, 0.0021]	+0.0007 [0.0002, 0.0012]
HaluEval	m2−m1	−0.0004 [−0.0020, 0.0012]	+0.0009 [0.0004, 0.0014]
HaluEval	m3−m1	+0.0001 [−0.0015, 0.0018]	+0.0010 [0.0004, 0.0016]
RAGTruth QA	m3−m0	+0.0076 [0.0045, 0.0109]	+0.0017 [−0.0270, 0.0291]
RAGTruth pooled	m3−m0	+0.0451 [0.0348, 0.0554]	+0.0962 [0.0644, 0.1295]

5.5. Explanation, Efficiency, and Judges

The explanation results use the fixed-hyperparameter m3 seed-42 checkpoint. Figure 9 ranks the top-15 features by mean absolute SHAP over 1,000 stratified HaluEval test rows. Context overlap dominates, followed by word count, question-answer cosine, character count, and question-answer Jaccard; the claim-contradiction mean enters at rank eight. The top ten features carry 77.9% of the total attribution mass, and exact TreeSHAP reproduces the model margin to 9.2×10^{-6}. The attributions are on the raw margin, not the display score. The m3 ranking agrees with the frozen B2 ranking (Spearman $\rho = 0.878$, Kendall $\tau = 0.730$), so the claim features refine the explanation rather than replace the original drivers. Cross-seed Spearman is 0.972 and 0.950, with nine of ten top features shared and an identical top-5 set. The explanation is stable to the seed but less stable to content edits. Table 14 shows the 20-case perturbation pilot: content edits move the score, structural edits barely move it, and the top-1 feature flips in 54.5% of entity perturbations and none of the other operators. The entity flip rate is higher than in B2, so the explanation is less stable when the answer

entity changes; per-operator n is small. A structured AI-assisted audit assessed 26 error-enriched cases selected under B2 and re-scored under m3, with each explanation checked against the answer-context evidence rather than the dataset label. Thirteen explanations were judged implausible, three unsure, and ten plausible, so the explanations are diagnostic aids, not faithful attributions. False positives inherit the length confound, and false negatives often reuse entities from the context. Because the sample is error-enriched, the percentages describe the audit, not the test set. The labels are author-checked pre-annotations: LLM tools assisted the annotation, and the authors reviewed and verified the final labels. Agreement with the pre-annotations was 0.98 for explanation, 0.94 for taxonomy, and 0.88 for claims. Cohen's κ was 0.965, 0.923, and 0.788; the authors changed 2%, 6%, and 12% of rows, and the blind Fleiss κ was 0.940, 0.868, and 0.761.

Table 14
Perturbation stability of the m3 raw score and SHAP ranking on HaluEval (seed 42, 20-case pilot).

Operator	n	Δscore	Large rate	Top-1 flip	Spearman
Entity replacement	11	0.339	0.364	0.545	0.848
Support removal	10	0.464	0.500	0.000	0.808
Number replacement	3	0.332	0.333	0.000	0.731
Date replacement	2	0.482	0.500	0.000	0.786
Irrelevant insert	20	0.003	0.000	0.000	0.922
Clause shuffle	2	0.000	0.000	0.000	0.999

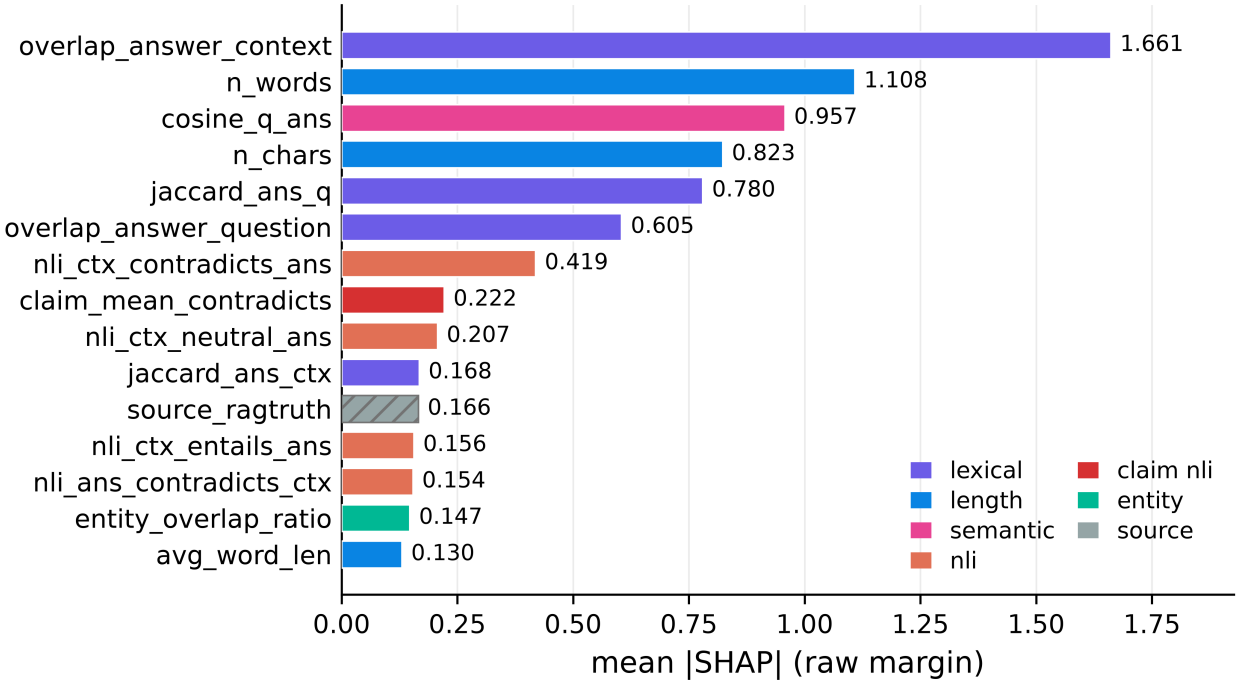


Figure 9: Mean absolute TreeSHAP for the fixed-hp m3 model (top 15 features).

Table 15 gives the component latency. Components are timed in separate instrumented passes over the same samples, so they do not sum to the end-to-end median, and retrieval and judge calls are excluded. NLI and NER dominate the base median, while XGBoost and SHAP are negligible. Adding the eight claim features raises the m3 path to 127.5 ms at the median. The same 200-row protocol on CPU takes 2.00 s per answer in a single contended run (p50 0.98 s, p95 7.56 s), so this figure depends on GPU feature extraction, and feature extraction, not the model, is the cost. Table 16 compares cost and latency: classical heads run in under a millisecond, and the two judges are 18 to 24 times slower than the m3 pipeline. The encoder alone takes 6.97 ms per row at batch 8 and fp16, so the feature

pipeline is about 18 times slower, but the encoder still needs a fine-tuned classification head and its training cost is not included. Model-only m3 inference is 0.008 ms, and the NLI-only head is the NLI step plus a linear head. Size is the seed-42 artifact on disk. The cost uses recorded token counts and an assumed $1.20/M output price, and the encoder timing is at 512 tokens, an upper bound for the 256-token baseline.

Table 15
Per-component latency in milliseconds (n = 200).

Component	p50	p95	mean
Core lexical	0.10	0.15	0.11
Entity (spaCy NER)	21.13	47.24	25.34
NLI (2 directions)	27.09	68.60	40.09
Semantic (MiniLM)	9.25	15.55	12.56
XGBoost inference	1.26	8.80	2.62
SHAP explanation	1.91	15.52	3.65
Base total	61.77	—	—
Claim NLI (m3)	43.75	83.21	53.11
Total m3	127.51	234.59	145.89

Table 16
Per-baseline cost and latency.

Baseline	Inputs	Size	p50 (ms)	Cost/1K
Heuristic (overlap)	text	—	0.009	≈$0
Logistic regression	26 feats	1.1 KB	0.001	≈$0.001
Random forest	26 feats	8.4 MB	0.331	≈$0.001
NLI-only	6 NLI probs	—	—	≈$0.001
TF-IDF (answer/full)	text	0.58/4.1 MB	0.014/0.110*	≈$0.001
XGBoost (conventional)	26 feats	0.49 MB	0.005	≈$0.001
RoBERTa-base	512 tokens (timing)	125M params	6.97	training only
EC-XGB (m3)	35 feats	0.68 MB	127.51	≈$0.001
GPT 5.6 Luna judge	API	—	2283.07	$0.234
Qwen 3.5 4B judge	4-bit local	4B params	3049.96	$0

LLM judges are not fair evaluators, and their preferences can shift with answer order [38]. Table 17 compares EC-XGB with two judges on the RAGTruth QA rows at natural prevalence. GPT-5.6 Luna reaches AUROC 0.910 against 0.558 for EC-XGB, and Qwen 3.5 4B also outperforms it, so both judges are better QA detectors and the feature model's advantage is cost. The target-fitted length-only baseline also ranks the QA rows better than the full feature model. Under matched multi-source training, the adapted encoder reaches 0.694 on the pooled RAGTruth test set, above EC-XGB's 0.582, so the encoder is stronger there. The feature model is less degraded on unseen tasks, but both are near chance. The length-matched panel changes the ranking: on the exact-length set, EC-XGB matches the stronger judge on AUROC and leads on F1, so the natural-prevalence gap is a domain-shift result rather than a length effect. The paired interval for GPT minus EC-XGB is −0.052 [−0.127, 0.024], and the AUROC interval is +0.008 [−0.034, 0.048]. Both include zero. Prompt choice moves GPT's F1 from 0.814 to 0.854 across three prompts, about the size of the model gap. GPT uses default decoding and three seeds, while Qwen is a single deterministic run, so the judge ranking is not stable.

5.6. Error Analysis and Discussion

On the seed-42 fixed-hyperparameter run, m3 makes 43 errors on the HaluEval test set (Figure 10), an error rate of 1.4%. The WP5 audit covers seven of the eight false positives and 12 of the 35 false negatives, chosen to cover the codebook. A separate 50-case sheet, reviewed by one author, adds 31 artifact cases and 19 content cases across 12 categories. These describe the reviewed cases, not the 3,000-row test population. Table 18 shows that length is a near-perfect label indicator on HaluEval, with false positives concentrated in the five-to-eight bin and false negatives

Table 17
Judge stress tests on the 900 RAGTruth QA rows and the 158-row length-matched HaluEval set.

Model	P	R	F1	AUROC	Flagged	p50 (ms)	Cost/1K
GPT 5.6 Luna judge	0.404	0.942	0.565	0.910	41.4%	2283	$0.234
Qwen 3.5 4B judge	0.570	0.638	0.602	0.778	19.9%	3050	$0
EC-XGB (m3)	0.183	0.998	0.310	0.558	96.8%	127.5	≈$0.001
Length-only	0.439	0.181	0.257	0.712	7.3%	—	≈$0
All-positive	0.178	1.000	0.302	—	100.0%	—	≈$0
Length-matched HaluEval, 158 rows, 50% positive							
GPT 5.6 Luna judge	0.945	0.798	0.865	0.957	42.2%	2283	$0.234
Qwen 3.5 4B judge	1.000	0.696	0.821	0.863	34.8%	3050	$0
EC-XGB (m3)	0.954	0.865	0.907	0.956	45.4%	127.5	≈$0.001
Length-only	0.515	0.443	0.476	0.521	43.0%	—	≈$0

in the short bins, although the 17+ bin has only two negative rows. The controls change the length distribution in two ways. A reweighting control equalizes six length bins with 247 examples each, and a length-matching control pairs one positive and one negative at each length. Reweighting gives EC-XGB a gain of 0.039 F1 and 0.023 AUROC over the length-only regression, which is already competitive on its own (Table 5). Length matching gives 79 pairs (158 examples) over 1,000 bootstraps. EC-XGB holds AUROC 0.956 while the length-only baseline falls to 0.521. The AUROC point gap is 0.435 (bootstrap mean 0.436 [0.351, 0.522]), and the F1 point gap is 0.431 (bootstrap mean 0.440 [0.334, 0.552]).

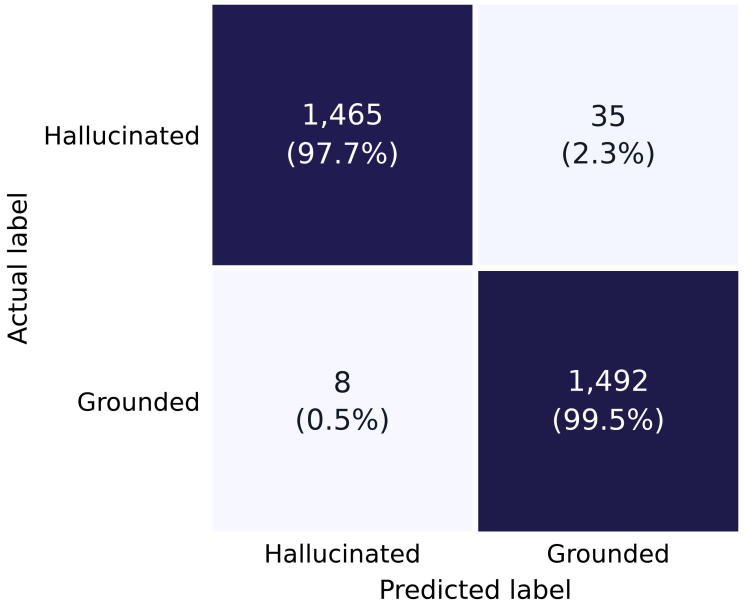


Figure 10: Confusion matrix for the seed-42 fixed-hp m3 model on the HaluEval test set at the 0.5 threshold (n = 3,000). Cell values are counts, with row-normalized percentages in parentheses.

Figure 11 shows the same pattern. Complete sentences are long, so the conventional model reads them as hallucinations, and the in-domain model is mostly a length and overlap detector. Published HaluEval QA results use different splits, supervision, and evidence access. A supervised 13B agent reports F1 0.724 on 100 rows, so these published numbers are context rather than a head-to-head bar. The benchmark also asks for hallucinated answers about five words longer, and the observed gap is 8.5 words, so any comparison that ignores this construction artifact overstates semantic detection. The claims in this paper are bounded by the tests that were not run. Retrieval quality

Table 18
HaluEval test errors of the conventional B2 model by answer length (counts seed 42, rates three-seed means).

Words	Rows	Hallucinated rate	FP rate	FN rate	Mean score
1	524	0.021	0.004	0.182	0.024
2	514	0.033	0.000	0.176	0.035
3–4	474	0.127	0.002	0.178	0.126
5–8	641	0.902	0.106	0.031	0.888
9–16	600	0.982	0.000	0.003	0.980
17+	247	0.992	0.333	0.000	0.993

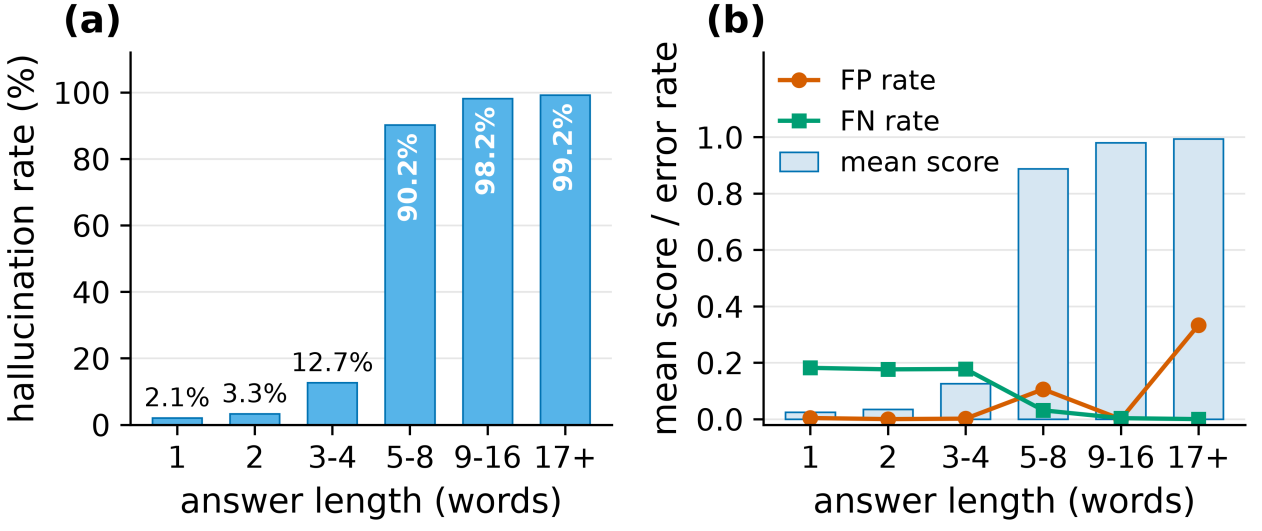


Figure 11: Hallucination rate, mean score, and error rates by answer length for the B2 conventional model.

and end-to-end verdict accuracy are not evaluated, the judge comparison controls length but not domain, and the audit is error-enriched. The deployed display calibrator was selected in an earlier fit that compared Platt and isotonic on the test split, and Table 8 refits Platt without that selection.

6. Conclusion

This study tested EC-XGB, a 35-feature evidence-consistent model, as a hallucination risk score and asked how its calibration and ranking change under domain and task shift. Its ranking survives a change of corpus, so the score can order answers for review, and the per-claim verdicts give a reviewer a reason for each flag. Its probability does not survive that change: the display score needs labeled target rows, and the target prevalence then sets its level. EC-XGB is therefore a domain-specific screening score, not a general-purpose detector. The next test is adaptation to an unseen QA domain, with thresholds selected under prevalence shift. A seen-task source ablation would then separate the RAGTruth rows from the source indicator.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing. **Saiful Alam Sabbir:** Data curation, Validation, Writing - review & editing. **MD. Miraz Ahamed:** Data curation, Validation, Writing - review & editing. **Ohidujjaman Tuhin:** Supervision, Validation, Writing - review & editing.

Declarations

- **Funding:** This research received no external funding.
- **Conflicts of Interest:** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Data availability:** Full raw datasets are not redistributed. For HaluEval (MIT), RAGTruth (MIT), and FaithBench (CC BY-NC-SA 4.0), the repository includes download scripts and a tracked license manifest with the source revisions and SHA-256 hashes, together with a 26-case audit file with benchmark excerpts. MedHallu (MIT) and the HaluBench subset (CC BY-NC 2.0) were obtained from their official releases. Splits and feature matrices can be reproduced from the original dataset sources. Model artifacts and the experiment manifest are available from the corresponding author on reasonable request.
- **Code availability:** The training pipeline, FastAPI inference service, and Next.js dashboard are available at `https://github.com/Atik203/HaluRISC` (branch `main`). The code is released under the MIT license.
- **Ethical approval:** No human-subject data were collected. The authors checked LLM-assisted pre-annotations of public benchmark rows for internal verification; this does not constitute human-subjects research, so ethics approval was not required.

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